

Graduation Project Proposal

**E-COMMERCE SALES AND CUSTOMER BEHAVIOR
ANALYSIS**

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1. INTRODUCTION

The rapid growth of e-commerce platforms has led to the generation of massive amounts of sales and customer data. Organizations need effective analytical techniques to transform this raw data into meaningful business insights. This project focuses on analyzing e-commerce sales and customer behavior to support data-driven decision-making and improve business performance.

2. PROBLEM STATEMENT

Many e-commerce organizations face challenges such as:

- Unclear sales trends and seasonal patterns
- Difficulty identifying profitable and low-performing products
- Limited understanding of customer purchasing behavior
- Low customer retention and engagement rates
- Lack of interactive dashboards for decision-making

3. OBJECTIVES

The main objectives of this project are:

- Analyze overall sales performance
- Identify top-selling and weak-performing products
- Segment customers based on purchasing behavior
- Measure customer retention and repeat purchase rates
- Apply advanced analytical techniques such as RFM and Cohort Analysis
- Design interactive dashboards for data visualization
- Provide business recommendations to improve sales and customer satisfaction

4. DATASET

The project dataset includes e-commerce Orders and Customer data with attributes such as: Order ID / Product Name / Quantity / Unit Price / Order Date / Customer ID / Country / Sales Revenue

The dataset may also include transaction history and customer activity records.

5. METHODOLOGY

The project will follow these steps:

5.1 Data Cleaning

- Handling missing values
- Removing duplicates
- Correcting inconsistent data formats

5.2 Exploratory Data Analysis (EDA)

- Sales trend analysis

- Product performance analysis
- Customer purchasing behavior analysis

5.3 Advanced Analytics

- RFM Analysis (Recency, Frequency, Monetary)
- Cohort Analysis
- Customer Segmentation

5.4 Data Visualization

- Building interactive dashboards
- Visualizing KPIs and business metrics

6. TOOLS

Excel for cleaning and preprocessing

Power BI for dashboards

Tableau for visualization

7. EXPECTED OUTCOMES

The expected outcomes of this project include:

- Interactive dashboards for sales and customer insights
- Identification of high-value customers and profitable products
- Detection of weak-performing areas
- Actionable recommendations for improving sales and retention
- Enhanced understanding of customer behavior patterns

8. CONCLUSION

Data analysis plays a vital role in improving business strategies and decision-making processes. This project aims to demonstrate how e-commerce data can be transformed into valuable insights that help organizations increase profitability, improve customer retention, and optimize overall performance.